

## Foreign Exchange Markets, at a glance.

A condensed, exam-ready reference for the whole module — the market, exchange rates, quotations and the arithmetic, market making, forwards, international payments and FX risk. Everything here comes from Novia's own study guide and class deck, reduced to the **definitions, tables, directions and traps** you are most likely to be tested on.

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## \$1.1 · INTRODUCTION — WHAT THE FX MARKET IS

**The foreign exchange market** — also known as **forex, FX or the currencies market** — is an **over-the-counter (OTC) global marketplace that determines the exchange rate for currencies around the world**. Participants **buy, sell, exchange and speculate** on the relative exchange rates of various currency pairs.

**EXAM WATCH** Currencies are always traded in pairs, so the “value” of one of the currencies in that pair is **relative to the value of the other**. There is no absolute price for a currency — only a price against another currency.

### Who makes up the market

Banks · forex dealers · commercial companies · central banks · investment management firms · hedge funds · retail forex dealers · investors.

### Scale

By far the largest financial market in the world — over **US\$5.2 trillion a day**. Made up of a **global network of financial centres that transact 24 hours a day, closing only on the weekends**.

## Free float vs fixed float

The value of a country's currency depends on whether it is a “**free float**” or a “**fixed float**”.

| REGIME               | DEFINITION                                                                                                                               | EXAMPLES                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Free-floating</b> | Relative value <b>determined by free-market forces</b> , such as supply-demand relationships.                                            | <b>US dollar · Japanese yen · British pound</b> |
| <b>Fixed float</b>   | A country's <b>governing body sets its currency's relative value to other currencies</b> , often by <b>pegging it to some standard</b> . | <b>Panamanian balboa · Saudi riyal</b>          |

**THE 24-HOUR MARKET** One of the most unique features of the forex market is that it is a global network of financial centres **transacting 24 hours a day, closing only on weekends**. As one **major forex hub closes, another hub in a different part of the world remains open for business**. This **increases the liquidity available in currency markets**, which adds to its appeal as **the largest asset class available to investors**.

## §1.2 · THE THREE TYPES OF FOREIGN EXCHANGE MARKET

| MARKET                      | DESCRIPTION                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Spot forex market</b>    | The <b>immediate exchange of currencies at the current exchange rate</b> — “ <b>on the spot</b> ”. Makes up a <b>large portion of the total forex market</b> and involves <b>buyers and sellers from across the entire spectrum of the financial sector</b> , as well as individuals exchanging currencies. |
| <b>Forward forex market</b> | An <b>agreement between the buyer and seller to exchange currencies at an agreed-upon price at a set date in the future. No exchange of actual currencies takes place, just the value.</b> Often used for <b>hedging</b> .                                                                                  |
| <b>Futures forex market</b> | Like the forward market — an agreed price at an agreed date. <b>The primary difference is that the futures market is regulated and happens on an exchange, which removes the risk found in other markets.</b> Also used for hedging.                                                                        |

### Advantages of the FX market

- **Fewer rules** than other markets — investors are not held to the strict standards or regulations found elsewhere
- **No clearinghouses and no central bodies** that oversee the forex market
- Most investors **will not pay the traditional fees or commissions** they would on another market
- **Open 24 hours a day**, so you can trade at any time — **no cutoff time** to participate, except heading into the weekend
- You can **get in and out whenever you want**, and buy as much currency as you can afford based on your account balance and your broker's rules for **leverage**

### Disadvantages of the FX market

- Being **unregulated also creates risks** — there is **no significant oversight that can ensure risk-free transactions**
- **Leverage magnifies profits but can also lead to high losses**; with **no set limits on leverage**, investors stand to lose a tremendous amount if trades move the wrong way
- Unlike stocks (dividends) and bonds (interest), **FX transactions rely solely on appreciation**, so they have **fewer residual returns** than some other assets
- **Lack of transparency** — a trader does not have full control over how trades are filled, may not get the best price, and may have a **limited view of information such as quotes**

**EXAM WATCH** The advantages and disadvantages are **two sides of the same fact**: the market's light regulation is simultaneously its biggest attraction and its biggest risk. Note too that the **futures** market is the exception — it is **regulated and exchange-traded**, which is precisely what removes the risk found in the other two.

### §1.2.1 · THE ROLE OF THE FX MARKET AND WHAT MOVES IT

**The role** — the foreign exchange market is a **24-hour global, decentralised, over-the-counter financial market for buying and selling foreign exchange**. It is the **largest and most liquid financial market in the world, with over US\$5.2 trillion a day**.

#### The five principal drivers behind foreign exchange activity

- The need to **pay for goods and services**, or to **execute foreign direct investments**
- **Global cross-border lending and investing** — for example buying or issuing foreign currency denominated bonds or equities
- **Hedging commercial or asset portfolio currency risks**
- **Direct investment in currencies as an asset class**
- **Currency speculation and arbitrage**

#### Factors that influence the foreign exchange market

The FX market is **very volatile**. The guide names eight influences:

| FACTOR                                                       | EFFECT                                                                                                                                                               |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inflation</b>                                             | <b>Devalues a currency because it decreases the currency's buying power.</b>                                                                                         |
| <b>Interest rates</b>                                        | Rates set by central banks affect the <b>cost of borrowing and the yield of holding a currency</b> . <b>Higher interest rates can lead to currency appreciation.</b> |
| <b>Political stability</b>                                   | <b>Stable political conditions encourage foreign investment</b> , which leads to appreciation.                                                                       |
| <b>Government debt</b>                                       | A country's <b>debt rating</b> can influence its currency exchange rate.                                                                                             |
| <b>Imports and exports</b>                                   | When exports exceed imports it creates a <b>trade surplus</b> , which <b>strengthens the economy and increases the value of the currency</b> .                       |
| <b>Market sentiment</b>                                      | Traders may trade on <b>beliefs about a currency's direction</b> , which increases or decreases demand.                                                              |
| <b>Economic growth</b>                                       | <b>Strong economic data</b> — GDP growth, inflation, employment and interest rates — can lead to <b>currency appreciation</b> .                                      |
| <b>Natural disasters, wars or technological advancements</b> | These events <b>can have a powerful impact on the foreign exchange market</b> .                                                                                      |

## §1.2.2 · PLAYERS IN THE FOREIGN EXCHANGE MARKET

The FX market is a global marketplace where currencies are exchanged against each other. Participants use it for a variety of reasons, including **hedging against currency and interest rate risk, diversifying portfolios, and speculating on geopolitical events.**

| PLAYER                           | ROLE — LEARN THESE TAGS VERBATIM                                                               |
|----------------------------------|------------------------------------------------------------------------------------------------|
| <b>Central banks</b>             | Considered <b>the market's stabilizers</b>                                                     |
| <b>Commercial banks</b>          | Considered <b>the core of the market</b>                                                       |
| <b>Hedge funds</b>               | Known for their <b>strategy expertise</b>                                                      |
| <b>Sovereign wealth funds</b>    | Considered <b>the silent giants</b>                                                            |
| <b>Forex dealers</b>             | <b>Provide access to the market</b>                                                            |
| <b>Corporations</b>              | <b>Practical users of the market</b>                                                           |
| <b>Proprietary trading firms</b> | Known for <b>taking risks</b>                                                                  |
| <b>Brokers</b>                   | <b>Facilitate transactions</b> in the market                                                   |
| <b>Liquidity providers</b>       | Considered <b>the backbone of the market</b>                                                   |
| <b>Speculators</b>               | Known for <b>moving prices</b>                                                                 |
| <b>Individual traders</b>        | A <b>small portion of the market</b> , but often use it for <b>speculation and day trading</b> |

**EXAM WATCH · THE SOUTH AFRICAN OVERLAY** Foreign exchange **futures** are traded on an exchange as **standardised contracts**. The South African stock exchanges — **JSE Limited, A2X Markets, ZAR X, CTSE (Cape Town Stock Exchange) and EESE (Equity Express Securities Exchange)** — trade **only in currency derivatives such as currency futures**. Most banks in SA have been registered as **Authorised Dealers (AD)**, trading in forex, by the Treasury department of **SARB**, and the list of ADs is in **Section A.2(A) of the Currency and Exchange Manual for Authorised Dealers**. The **FSCA is the market regulator in South Africa that oversees forex trading** — trading with FSCA-regulated brokers ensures a high level of security.

### The 24-hour session handover

While most trading in a currency occurs when its **main forex trading market is open**, many other banks around the world hold foreign currencies, **enabling them to continue trading when the main market is closed**. Dealers are in constant contact and trading happens continuously on a 24-hour basis: **as the Asian session ends the European session begins, followed by the North American session, and then back to the Asian session.**

Example: **trading runs from 22:00 GMT on Sunday (Sydney) until 22:00 GMT Friday (New York)**. GMT — **Greenwich Mean Time** — means what time it is anywhere in the world at a certain point. If it is about **07h00 in SA** it is around **17h00 in Sydney** on the same day.

### §1.2.3 · PURCHASING POWER PARITY (PPP)

**Purchasing power parity** — a popular **macroeconomic analysis metric used to compare economic productivity and standards of living between countries**. It compares different countries' currencies through a “**basket of goods**” approach: PPP is **the exchange rate at which one nation's currency would be converted into another to purchase the same amount of a large group of products** — the rate at which **the same products and services can be purchased in each country**.

#### The four things to know

- PPP is a metric used by macroeconomic analysts comparing currencies through a **basket of goods**
- It allows economists to **compare economic productivity and standards of living between countries**
- **Some countries adjust their gross domestic product (GDP) figures to reflect PPP**
- **Some feel that PPP does not reflect reality**, due to differences in **local costs, taxes, tariffs and competition**

**EXAM WATCH** The PPP exchange rate is used to convert the local currency of a target nation into a common currency applicable to all nations. Normally this common currency is the **US dollar**, or what is called the **international dollar** — a currency designed to serve as a baseline.

#### Worked example — the guide's own

Suppose it costs **\$10** to buy a shirt in the U.S., and **€8.00** to buy an identical shirt in Germany. To make an apples-to-apples comparison, first convert the €8.00 into U.S. dollars. If the exchange rate is such that the shirt in Germany costs **\$15.00**:

$$\text{PPP} = 15 \div 10 = 1.5$$

In other words, **for every \$1.00 spent on the shirt in the U.S., it takes \$1.50 to obtain the same shirt in Germany when buying it with the euro**.

## §1.2.4 · FACTORS THAT INFLUENCE EXCHANGE RATES AND INVESTOR CONFIDENCE

| FACTOR                     | MECHANISM                                                                                                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interest rates</b>      | Central banks set interest rates, which <b>affect the cost of borrowing or the yield of holding a currency</b> . <b>Higher interest rates can lead to a stronger currency, while lower interest rates can lead to depreciation.</b> |
| <b>Political stability</b> | <b>Enhances investor confidence</b> , which can lead to a country's currency being <b>perceived as a safe store of value</b> . <b>Political instability, corruption or economic uncertainty can deter investors.</b>                |
| <b>Market sentiment</b>    | <b>Investor confidence can significantly impact exchange rate fluctuations</b> . If investors view a country's economy as stable, <b>they may be more likely to invest in its currency.</b>                                         |
| <b>Trade balances</b>      | A country's trade balance can impact the value of its currency. <b>If a country imports more goods and services than it exports, this can lead to inflation and decrease the value of its currency.</b>                             |
| <b>Economic growth</b>     | A country's <b>net exports are inversely correlated with the strength of its domestic currency.</b>                                                                                                                                 |
| <b>Other factors</b>       | <b>Inflation · currency reserve status · speculation · public debt · unemployment rates.</b>                                                                                                                                        |

**WATCH THE TWO LISTS** The guide gives an **eight-item list at §1.2.1** (factors that influence the FX market) and a **six-item list at §1.2.4** (factors that influence exchange rates and investor confidence). They **overlap but are not identical** — §1.2.4 adds **trade balances** and **currency reserve status**, and frames economic growth through **net exports**. If a question cites a section, answer from that section's list.

**EXAM WATCH — THE ONE THAT READS BACKWARDS** **“A country's net exports are inversely correlated with the strength of its domestic currency.”** A **stronger** domestic currency makes exports **more expensive abroad**, so net exports **fall**. Do not confuse this with the trade-balance line, which runs the other way — a **trade surplus strengthens** the currency. Cause and effect point in opposite directions in the two statements, and both are examinable.

### § 1.3.1 · EXCHANGE RATES — THE CONCEPT AND HOW IT ALL BEGAN

The foreign exchange market is an **over-the-counter global market where the buying and selling of global currencies occurs, determining their exchange rates**. It is made up of **banks, forex dealers, commercial companies, central banks, investment management firms, hedge funds, retail forex dealers and investors that all trade currency pairs**.

#### The timeline — high-yield dates

| WHEN            | WHAT HAPPENED                                                                                                                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Antiquity       | As long as humans have been trading there has been an FX market. <b>Ancient civilisations traded goods and currencies through metal coins, whose value was based on their weight.</b>                                                                       |
| ≈ 500 years ago | The <b>first true forex market was in Amsterdam</b> . The exchange allowed people to <b>freely trade currencies to stabilise exchange rates</b> .                                                                                                           |
| 1875            | The <b>gold standard was implemented</b> — countries were <b>only allowed to print currency equal to the amount of their gold reserves</b> . Gold was the metal of choice because it is <b>rare, malleable, tough to corrode and hard to obtain</b> .       |
| 1903 → 1913     | Forex trading firms in London grew from <b>three to 71</b> .                                                                                                                                                                                                |
| The world wars  | The gold standard <b>could not hold up</b> , because countries had to <b>print more money to finance expenses</b> .                                                                                                                                         |
| After WWII      | The <b>Bretton Woods system</b> was established — most currencies <b>pegged to the U.S. dollar</b> , which was itself <b>backed by gold reserves</b> .                                                                                                      |
| 1971            | <b>President Nixon announced a freeze on the dollar's convertibility to gold</b> , due to rising inflation and a possible gold run.                                                                                                                         |
| 1973            | The <b>gold standard was completely abolished</b> ; the dollar was <b>no longer backed by gold</b> and FX <b>switched to a free-floating system</b> — currencies free to peg to any currency, or remain unpegged and let supply and demand determine value. |
| 1992            | The <b>Maastricht Treaty</b> was signed to introduce a single currency in Europe.                                                                                                                                                                           |
| 1 January 1999  | The <b>EUR was launched</b> in a number of European countries.                                                                                                                                                                                              |

**EXAM WATCH** The three dates that carry most of the marks: **1875 gold standard in · 1971 Nixon freezes convertibility · 1973 gold standard abolished and FX goes free-floating**. Note the pair: the **US dollar became fiat money in 1971 when the country left the gold standard**.

## The three exchange rate regimes

| REGIME                                         | DEFINITION                                                                                                 |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Floating exchange rates</b>                 | Price determined <b>only by demand and supply of the currency</b> — <b>no government intervention</b> .    |
| <b>Fixed exchange rates</b>                    | The value of a currency <b>fixed in relation to an anchor currency</b> — <b>not allowed to fluctuate</b> . |
| <b>Dirty floating or managed exchange rate</b> | Rate <b>influenced by government via the central bank</b> around a preferred rate.                         |

## Money vs currency

**Money** — the **broader term**: an **intangible system of value that makes the exchange of goods and services possible, now and in the future**. **Currency** is simply one tangible form of money.

### The two functions the guide names

**Store of value** — money has and maintains a certain value that supports ongoing exchanges; people know the money received today will essentially have the same value next week.

**Unit of account** — it can be used to **account for changes in the value of items over time**. Businesses use money as a unit of account when they **prepare a budget or give assets a value**, and **profits and losses are established and relied upon using money as a unit of account**.

### The five properties of money

- **Fungible** or exchangeable — it does not need to be re-valued for every transaction
- **Durable** — it lasts for many exchanges over time
- **Convenient** to carry and divide
- **Recognisable**, so people can trust it and confidently complete exchanges
- **Stable supply**, so that its value is reliable

Currency is a **form of money used every day by people all over the world**. **Cheques are another form of money, known as money substitutes**. **Cigarettes have even been a form of money**, as they were for soldiers during the Second World War.

### §1.3.4–1.3.5 · FIAT VS REPRESENTATIVE MONEY VS CRYPTO

The main difference between fiat currency and representative money is that **fiat currency is backed by the government's trust, while representative money is backed by a commodity.**

|                  | FIAT CURRENCY                                                                                                                                                                      | REPRESENTATIVE MONEY                                                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Backed by</b> | The <b>government's trust. Not backed by a commodity with intrinsic value</b> — the <b>government's official recognition of it as a medium of exchange is what gives it value.</b> | A <b>commodity, such as gold or silver</b> , and <b>can be exchanged for that commodity.</b>                                                                                                              |
| <b>Form</b>      | <b>Coins, banknotes or digital currencies</b> — physical money, paper or coins.                                                                                                    | A <b>cheque or other form of currency that can be exchanged for physical money in a stated amount</b> ; also a form of currency that <b>represents the intent to pay, such as a cheque or debit card.</b> |
| <b>Example</b>   | The <b>US dollar became fiat money in 1971</b> when the country left the gold standard.                                                                                            | <b>The gold standard</b> — each country's currency tied to a fixed amount of gold.                                                                                                                        |

Both fiat and representative money possess the value they claim to have. Fiat money is backed by the government that issues it.

### Fiat vs cryptocurrency

|                   | FIAT                                                                                                 | CRYPTOCURRENCY                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Validation</b> | A <b>bank validates the transaction.</b>                                                             | <b>Does not need an intermediary</b> to validate a transaction — it originated as a <b>peer-to-peer electronic cash system</b> and is <b>secured by cryptography.</b> |
| <b>Control</b>    | Controlled by a <b>government authority or central bank.</b>                                         | <b>Not controlled</b> by a government authority or central bank.                                                                                                      |
| <b>Supply</b>     | <b>More fiat money can be printed at any time</b> — in theory <b>no limits on supply.</b>            | <b>Fixed supply</b> — most cryptocurrencies create a <b>set amount of coins that can ever exist.</b>                                                                  |
| <b>Volatility</b> | Comparatively stable.                                                                                | <b>More volatile than fiat</b> , because it is <b>not regulated or backed by a government.</b>                                                                        |
| <b>Storage</b>    | Can be represented in <b>physical or digital form</b> , so it can be <b>stored in various forms.</b> | <b>Cannot</b> be represented with physical notes and coins — <b>must be stored in digital, or cryptocurrency, wallets.</b>                                            |

## Factors in the supply and demand of a currency

| FACTOR                      | MECHANISM                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------|
| International trade         | When a country exports more than it imports, the demand for its currency increases. |
| Inflation                   | Inflation is the rate at which a currency loses its purchasing power.               |
| Economic growth             | A country's economic prosperity determines its population's purchasing power.       |
| Interest rates              | In a growing economy the demand for money increases, which raises interest rates.   |
| Domestic economic downturns | When a currency is not in demand, its value will fall relative to other currencies. |
| Other factors               | Capital flow and money supply.                                                      |

The most common method to value currency is through exchange rates. Supply and demand for any given currency, and thus its value, are **not influenced by any single element but by several**, which generally fall into three categories: **economic factors**, **political conditions** and **market psychology**.

## Currency appreciation and depreciation

|                  | CURRENCY APPRECIATION                                            | CURRENCY DEPRECIATION                                            |
|------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| Definition       | A currency's value <b>increases</b> relative to another currency | A currency's value <b>decreases</b> relative to another currency |
| Effect on prices | <b>Lower effective price for imported goods</b>                  | <b>Higher import prices</b>                                      |

**EXAM WATCH** “There is no optimal value for a currency.” A stronger currency **does not necessarily benefit everyone in the economy** — it **benefits some people and sectors but hurts others. It is therefore not necessarily better; it is just different.** When a currency rises so that it exchanges for more of other currencies we say it “**appreciates**” or “**strengthens**”; when the rate falls and it trades for less we say it “**depreciates**” or “**weakens**”.

## §1.4 · WHO WINS FROM A STRONG RAND — AND WHO LOSES

| TYPE OF TRANSACTION                        | IMPACT OF A STRONG RAND | IMPACT OF A WEAK RAND |
|--------------------------------------------|-------------------------|-----------------------|
| South African exporting company            | Less competitive        | More competitive      |
| South African importing company            | More competitive        | Less competitive      |
| SA traveller abroad                        | More competitive        | Less competitive      |
| Foreign traveller visiting SA              | Less competitive        | More competitive      |
| SA investor, investing abroad              | More competitive        | Less competitive      |
| An international investor, investing in SA | Less competitive        | More competitive      |

**THE PATTERN** Read it as one rule: **a strong rand favours whoever is spending rand abroad** — importers, SA travellers, SA investors going offshore. **A weak rand favours whoever is bringing foreign currency into South Africa** — exporters, foreign tourists, foreign investors. Every row in the table is an instance of that single idea.

### Effect of exchange rates on the economy

#### Import industries

**USD appreciates / ZAR depreciates** — e.g. USD/ZAR from 7.11 to 7.20 → **import prices increase** → **inflation increases**.

**USD depreciates / ZAR appreciates** — e.g. USD/ZAR from 7.07 to 6.98 → **import prices decrease** → **inflation decreases**.

#### Export industries

**USD depreciates / ZAR appreciates (7.07 → 6.98)** — prices of exports **increase: more expensive** (receive **less** rand per USD) → **levels of exports decrease** → **levels of employment decrease**.

**USD appreciates / ZAR depreciates (7.11 → 7.20)** — prices of exports **decrease: cheaper** (receive **more** rand per USD) → **levels of exports increase** → **levels of employment increase**.

**EXAM WATCH — THE SINGLE MOST IMPORTANT CONVENTION IN THE MODULE** Banks will use a buying rate when they are purchasing foreign currency from a customer and a selling rate when they sell foreign currency to a customer. Exchange rates are always from a bank's point of view.

### Bid and ask

The **bid price** is what the dealer is willing to pay (buy) for a currency; the **offer/ask price** is the rate at which a dealer will sell the same currency.

**SA importers will buy currency at a rate and the bank sells the currency at a rate.**

**SA exporters will sell currency at a rate and the bank buys the currency at a rate.**

### How the bank sets its rates

Local banks look at their own “**exchange position**”, use the **SARB base rate for USD**, and determine their position from the day's purchases and sales, **both spot and forward**, of foreign currencies. Banks have sophisticated systems to calculate **cross rates for currencies not quoted on the Reuters system against the rand**. This is done **every morning in the bank's treasury department before the branches can perform foreign transactions**.

**ISO currency codes** — **three-letter alphabetic codes that represent the various currencies used throughout the world**. When **combined in pairs they make up the symbols and cross rates used in currency trading**. Each country-specific three-letter code also has a **corresponding three-digit numeric code**. The codes are identified by the **International Organisation for Standardisation (ISO)**, a non-governmental organisation providing standards for manufacturing, commerce, technology and communication; for currencies the **governing document is ISO 4217:2015**.

### Base and quote

The **first three-digit code used to designate a currency in a currency pair quotation is called the base currency**, and the **second currency is called the quote currency**. The currency pair indicates how much of the quote currency is needed to purchase one unit of the base currency.

**Example.** EUR/USD is the quote for the euro against the U.S. dollar. A quoted price of **1.2500** means **one euro is exchanged for 1.25 U.S. dollars**, because EUR is the base currency and **USD is the quote (or counter) currency**. Another way of looking at it: **it will cost you \$125 (USD) to buy €100 (EUR)**.

EUR/USD — euro / U.S. dollar

GBP/USD — sterling / U.S. dollar

USD/JPY — U.S. dollar / yen

USD/CHF — U.S. dollar / Swiss franc

USD/ZAR — U.S. dollar / SA rand

In **USD/ZAR** the **USD** is the base currency and the **ZAR** is the quoted or second currency.

## §1.5 · THE SPOT EXCHANGE MARKET

**Spot market** — the place where **financial instruments are traded for cash for immediate delivery**. Assets traded include **commodities, currencies and securities**. **Delivery occurs when the buyer and seller exchange cash for the financial instrument**. A futures contract, by contrast, is based on delivery of the underlying **at a future date**.

### The five facts about spot markets

- **Financial instruments trade for immediate delivery** in the spot market
- **Many assets quote a spot price and a futures or forward price**
- **Most spot market transactions settle one business day after the transaction — but foreign exchange settles in two business days**
- Spot market transactions **can take place on an exchange or over the counter**
- Spot markets can be **contrasted with derivatives markets**, which instead trade forwards, futures or options contracts

**Spot rate transaction** — an **outright purchase or sale of one currency for another for delivery two working days after the dealing date** (the date the contract was established). **If the spot date falls on a public holiday in one or both centres of the two currencies involved, the value date will move to the next “good” working day.**

### Dealing day → settlement day

| DEALING DAY | SETTLEMENT DAY   |
|-------------|------------------|
| Monday      | <b>Wednesday</b> |
| Tuesday     | <b>Thursday</b>  |
| Wednesday   | <b>Friday</b>    |
| Thursday    | <b>Monday</b>    |
| Friday      | <b>Tuesday</b>   |

### The points to quote back

The spot rate offers **adaptability and up-to-the-minute pricing**. **Market demand and supply, trading activities, and the bid-ask spread are the three important trends determining the spot price**. It allows participants to transact **without a long-term commitment**, giving greater flexibility and reducing the risk of currency fluctuations. Should a customer request a rate **value today or value tomorrow**, the rate quoted **will be less favourable than the spot rate, due to interest rate differentials** — and **keep the current cut-off times in mind**. **One would not be able to transact value today when the beneficiary resides in Australia and payment is to be made in AUD from South Africa.**

## §1.5 · DIRECT AND INDIRECT QUOTATIONS — PRICE MAKER VS PRICE USER

When banks — the **price maker** — quote a price they indicate to the counterparty — the **price taker or price user** — where they are prepared to **buy (bid)** and/or **sell (offer)** the **base currency** against the **variable (quoted) currency**. The price user then “**selects**” to buy or sell the base currency at the quoted price.

### Direct quoted currency

Quoted with **US dollars as the base**. Expresses **the amount of local currency needed to buy or sell 1 US dollar**.

Example: **1 USD/ZAR 7.5050** — one US dollar equals R7.5050.

### Indirect quoted currency

Quoted with the **local or domestic currency as the base**.

Indicates **the amount of US dollars needed to buy or sell one unit of the local currency**. Example: **GBP/USD = 1.7575** — one British pound equals 1.7575 US dollars.

### Direct quotation worked through — 1 USD/ZAR 6.90 / 6.91

| DIRECT QUOTED CURRENCY                                    | BID — 6.9000                                            | OFFER — 6.9100                                          |
|-----------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| <b>Quoting bank (price maker)</b><br>USD base, ZAR quoted | <b>Buys \$</b> at 6.9000<br><b>Sells rand</b> at 6.9000 | <b>Sells \$</b> at 6.9100<br><b>Buys rand</b> at 6.9100 |
| <b>Price user (counterparty)</b><br>USD base, ZAR quoted  | <b>Sells \$</b> at 6.9000<br><b>Buys rand</b> at 6.9000 | <b>Buys \$</b> at 6.9100<br><b>Sells rand</b> at 6.9100 |

### Indirect quotation worked through — 1 GBP/USD 1.8525 / 1.8530

| INDIRECT QUOTED CURRENCY                                  | BID — 1.8525                                         | OFFER — 1.8530                                       |
|-----------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| <b>Quoting bank (price maker)</b><br>GBP base, USD quoted | <b>Buys £</b> at 1.8525<br><b>Sells \$</b> at 1.8525 | <b>Sells £</b> at 1.8530<br><b>Buys \$</b> at 1.8530 |
| <b>Price user (counterparty)</b><br>GBP base, USD quoted  | <b>Sells £</b> at 1.8525<br><b>Buys \$</b> at 1.8525 | <b>Buys £</b> at 1.8530<br><b>Sells \$</b> at 1.8530 |

**EXAM WATCH — THE RULE BEHIND BOTH TABLES** The price maker will buy the base currency low (USD/ZAR 6.9000 or GBP/USD 1.8525), at the “**smaller price**” or “**cheaper rate**”, and sell the base currency high (USD/ZAR 6.9100 and GBP/USD 1.8530), at the “**bigger price**” or “**more expensive price**”. The **spread is not a dealer profit but a “risk spread”**.

## Price decomposition

Consider bid and offer prices **USD/ZAR 6.9525 (bid)** and **USD/ZAR 6.9575 (offer)**. In the interbank market the price is written **6.9525/75**. When banks make a price to other banks they quote **25/75** — the “small figures”, where **25 is the bank’s bid price to buy the base currency** and **75 is the bank’s offer price to sell the base currency and buy the variable currency**. The 6.95 is not part of the price, as the assumption is made in the interbank market that the big figure (6.95) is known to all participants.

| QUOTE USD/ZAR                                    | BIG FIGURE   | BID — 25                                                                                          | OFFER — 75                                                                                        |
|--------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>6.9525 / 6.9575</b><br>(6.9525/75)            | <b>6.95</b>  | <b>Bank buys USD</b><br>Bank sells ZAR                                                            | <b>Bank sells USD</b><br>Bank buys ZAR                                                            |
| <b>GBP/ZAR 10.2525 / 10.2575</b><br>(10.2525/75) | <b>10.25</b> | <b>Bank buys GBP</b><br>Bank sells ZAR<br>cross-rate legs: buys GBP sells USD; buys USD sells ZAR | <b>Bank sells GBP</b><br>Bank buys ZAR<br>cross-rate legs: sells GBP buys USD; sells USD buys ZAR |

## Reciprocal rates

Any quotation with a particular currency as the base **can be converted into the equivalent quotation with that currency as the variable or quoted currency** — converting a direct quoted pair into an indirectly quoted pair and vice versa.

### The guide's worked example

USD/ZAR 6.8525 →  $1 \div 6.8525$  **0.145932...**

USD/ZAR 6.8575 →  $1 \div 6.8575$  **0.145826...**

**ZAR/USD quotation published by the guide** **0.1458 / 0.1460**

**EXAM WATCH** The smaller number is always on the left and the higher number on the right, so the two sides of the quotation are reversed. In either case the price maker will buy the base currency on the left against the variable currency and sell the quoted currency on the right.

**CROSS-RATES** A cross-rate is an exchange rate between two currencies, calculated from their common relationships with a third currency — namely the US dollar. When cross rates differ from the direct rates between two currencies, inter-market arbitrage is possible. By market convention the **euro, Australian dollar, pound sterling and New Zealand dollar are the base currency** in cross-rate quotations; for other pairs there is **no universal or agreed convention** on which currency is the base. **In South Africa all exchange rates are quoted against the US dollar, and all other currencies are cross rates.** The **USD/ZAR rate is quoted on Reuters**, and local banks calculate from the respective US dollar rate to obtain a cross rate. Internationally a cross rate is any quotation between two currencies **where the US dollar is not one of them**, e.g. **EUR/CHF**.

## \$1.5.5–1.5.6 · POINT'S VALUE AND THE ADVANTAGES OF SPOT RATES

**Points or “pips”** — the **smallest change in the price of a currency**. Different currency pairs are quoted with **different accuracy, or a different number of decimals**. **Most currencies are quoted with an accuracy of 0.0001, but others — such as the yen and its cross-rates — are quoted with an accuracy of 0.01**. The number of decimals is determined by **market convention**.

The value of one point **differs for different currencies**, and can differ **for the same currency depending on the size or amount of the transaction**.

$$\text{Value of one point} = \text{Size of deal} \times \text{Net change in rate (points change)}$$

By applying the formula, **the resulting value change will be expressed in the base currency**.

### Worked example

You have a spot position of **USD 100,000 in yen**. The USD/JPY rate changes from **114.21 to 114.20**. The change in the value of your position is:

$$\text{Value of one point} = \text{USD } 100,000 \times 0.01 = \text{USD } 1,000$$

If the position was for **USD 200,000** the value change would have been **\$2,000**.

## The six advantages of spot exchange rates

| ADVANTAGE                      | WHY                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Simplicity</b>              | Straightforward to understand — <b>simply the price at which one currency can be exchanged for another on the spot date</b> .                                              |
| <b>Flexibility</b>             | A flexible option for businesses needing to exchange currencies <b>short term</b> ; allows companies to <b>take advantage of favourable exchange rates as they arise</b> . |
| <b>Speed</b>                   | Transactions are <b>completed quickly, typically within two business days</b> — an advantage for companies needing to <b>pay for goods or services promptly</b> .          |
| <b>Cost-effectiveness</b>      | Generally cost-effective for businesses exchanging <b>small amounts</b> , as it <b>does not require any upfront payment or deposit</b> .                                   |
| <b>Real-time pricing</b>       | The rate <b>reflects current market conditions and is updated in real time</b> , allowing informed decisions.                                                              |
| <b>Transparent environment</b> | Spot markets facilitate trading <b>where transactions occur at prevailing prices that are public information</b> .                                                         |

## §1.6 · PRINCIPLES OF MARKET MAKING

There are **two main objectives in quoting a buy and sell price in any market**: firstly, to **create a marketplace where customers can buy or sell the particular item of their choice**; and secondly, to **make a profit**.

### Why a two-way price is needed at all

In an ordinary market a **customer only looks to buy an item and not to sell it**, so no two-price is needed — the trader offers the item at a specific price and the buyer buys at that price. **In the financial world a customer can be both a buyer and a seller of the same asset**: a client can **sell USD against the ZAR** (proceeds from exports) and can also **buy USD against the rand** to pay for imports of raw material or capital goods. This means the bank must make **both a buying (bid) price and an offer (selling) price**, as it is not always sure what the client is looking to do.

**TAKING A POSITION** It is **unlikely the market will deal on both prices simultaneously** under normal conditions. **Dealing on one or the other side of a two-way price results in the trader “taking a position”**, and his price must then reflect his views on current market conditions and future expectations. If he goes **“long”** (bought the base) he expects to **sell the base currency at a higher rate**; if he goes **“short”** (sold the base) he expects to **buy the base currency cheaper, at a lower price**.

### The seven factors a market maker weighs before taking a position

- What is the **current level of market price** being made
- What is the **current spread** — **the smaller the dealer spread, the more stable, “less volatile”, the market**
- What are my **profit/loss guidelines** — dealing limits for the **size of position** allowed and the **loss limit** before he must liquidate
- What are the **normal amounts traded** per currency pair quote
- What is the **liquidity of the market** — can large volumes be traded **without causing the market to react adversely**
- What is the **volatility of the market** — it **impacts the bid-offer spread**
- What are the **restrictions on whom I can deal with** — **credit lines and dealing limits between banks**; does he have enough counterparties to **“cover” his net open exposure**

**EXAM WATCH · APPRECIATION WITHIN A PAIR** A currency pair consists of a **base currency and a variable currency**. **If the base currency appreciates and gets stronger, the variable currency gets weaker** — i.e. you need **more of the variable currency** to buy the base. Example: **if USD/ZAR moves from 7.1025 to 7.1552, the base currency (USD) has appreciated and the variable currency (ZAR) has depreciated**, because you now need more rand to buy one unit of the base. If you are selling USD and the USD has appreciated, **you will receive more ZAR**.

## §1.6 · THE IMPACT OF SARB INTERVENTION ON MARKETS

|                                      | EXCHANGE RATE                                                         | M1                                            | RAND INTEREST RATES                                       |
|--------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| <b>SARB buys USD</b><br>+USD / -ZAR  | <b>USD: appreciate</b><br><b>ZAR: depreciate</b><br>From 6.95 to 7.10 | <b>Increase</b><br>(more ZAR in circulation)  | <b>Decrease</b><br>5% to 2%<br>(more ZAR in circulation)  |
| <b>SARB sells USD</b><br>-USD / +ZAR | <b>USD: depreciate</b><br><b>ZAR: appreciate</b><br>From 7.25 to 7.05 | <b>Decrease</b><br>(less rand in circulation) | <b>Increase</b><br>4% to 7%<br>(less rand in circulation) |

**SARB will buy USD to weaken (depreciate) the ZAR in order to...**

- **Help exports**
- **Increase capital inflows**, as investors will receive **more rand per 1 USD**
- **Gold and foreign exchange reserves will increase**

**SARB will sell USD to strengthen (appreciate) the ZAR in order to...**

- **Promote imports**
- **Gold and foreign exchange reserves will decrease**
- **Limited capital inflows**

### Spot vs forward exchange rates

| SPOT EXCHANGE RATES                                                          | FORWARD EXCHANGE RATES                                                     |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Refers to the <b>current rate</b> .                                          | Refers to the rate <b>at some point in the future</b> .                    |
| Provides <b>flexibility and real-time pricing</b> .                          | Provides <b>certainty and protection against currency fluctuations</b> .   |
| Exchange happens <b>on the spot date, typically within 2 business days</b> . | Exchange happens <b>at an agreed date in the future, e.g. 3–180 days</b> . |

**EXAM WATCH** The direction is counter-intuitive until you see it as a flow of rand: **when the SARB buys dollars it pays out rand, so more rand circulates — M1 rises and rand interest rates fall.** When it **sells dollars it takes rand in, so less rand circulates — M1 falls and rand rates rise.**

## §1.7 · FORWARD EXCHANGE CONTRACTS (FECS)

**Forward foreign exchange** — an **OTC market used to hedge or speculate in future foreign exchange rate movements**. The market determines a **future exchange rate based on the interest rate differential between the two currencies involved and their current (spot) exchange rate**. In the strictest sense of the word, **forward foreign exchange is a derivative of interest rates and the spot foreign exchange rates**.

**Outright forward** — an **outright purchase or sale of one currency in exchange for another for settlement on a fixed date in the future other than the spot value date**. The party agreeing to **buy** the base currency in the future assumes a **“long position”**; the party agreeing to **sell** assumes a **“short position”**. The price agreed is the **delivery price or contract price**, equal to the forward price at the time the contract is entered into — **the spot price adjusted for the interest rate differential between the two currencies for the specific period**.

### Why it exists — the guide's British manufacturer

A British manufacturer contracts today to deliver a chemical process plant in a year's time in the USA at a price **fixed in dollars today**. He **would not know what the sterling equivalent of his dollars would be** in one year, yet his **costs are reckoned in sterling** because he sources the plant in the UK before exporting it. He **determines his profit or mark-up on the current spot rate**, but is **not sure whether the rate will move against him and turn the profit into a loss**. He is **already exposed to operational risk** — labour and technical problems, wage costs, raw material and fuel prices, freight increases and delivery dates — **without also exposing himself to exchange risk**. The forward market lets him **sell his USD for GBP today at a pre-agreed rate for delivery in one year's time**, eliminating exchange rate risk.

**THE “RISK FREE” FORWARD RATE** When calculating forwards there are **only two physically traded instruments on which to base the calculation — the immediate cash markets for interest rates and exchange rates**. The market's calculation of derivative prices from these **does not include any element of profit or loss**; the theoretical calculation **exactly corresponds to the relationship of the rates used**. This is why it is known as the **“risk free” or “arbitrage free” forward rate**. The forward rate **ignores any outside pressure on exchange rates**.

**EXAM WATCH — THE TRADE-OFF** Once calculated and dealt, a forward **locks in a future price**, making it **particularly useful for traders or users of foreign currency who need to know exactly what their future costs will be**. But because the rate is **unaffected by movements in exchange rates over time**, while a forward position **guards against loss from adverse rate movements it also precludes profits from favourable movements**. With the advent of currency futures and options, forward FX is less a tool of speculators than before — one of its major uses is to arbitrage in the interest rate markets.

## §1.7.1–1.7.4 · PREMIUM VS DISCOUNT, SHORT DATES AND CONTRACT TYPES

**Premium vs discount** — a currency with lower interest rates will trade at a forward premium in relation to a currency with a higher interest rate. A premium is beneficial to the exporter and a cost to the importer, while the converse applies in the case of a discount.

|                 | <b>PREMIUM</b><br>SA INTEREST RATE IS <b>HIGHER</b> THAN<br>THE OTHER COUNTRY | <b>DISCOUNT</b><br>SA INTEREST RATE IS <b>LOWER</b> THAN<br>THE OTHER COUNTRY |
|-----------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Importer</b> | <b>PIL — Premium Importer Loss</b>                                            | <b>DIG — Discount Importer Gain</b>                                           |
| <b>Exporter</b> | <b>PEG — Premium Exporter Gain</b>                                            | <b>DEL — Discount Exporter Loss</b>                                           |

Because there is a **wide range of interest rates applying in foreign countries, some higher and some lower**, both cases arise. **SA NCD rates are mostly higher than SOFR, and in some cases lower**. The mnemonic to memorise is **PIL · PEG · DIG · DEL**.

### §1.7.2 Short dates

**Value dates shorter than one month are referred to as short dates** and are run by banks as a separate position. Forward transactions with a value of less than one month are “**short dates**”, and the terminology is the same as for money market deposits:

| <b>TERM</b>        | <b>DEFINITION</b>                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------|
| <b>Overnight</b>   | A deposit or FX swap <b>from today until tomorrow</b> — from time zero (T0) to today plus one (T1). |
| <b>Tom-next</b>    | A deposit or FX swap <b>from “tomorrow” to the next day (spot)</b> — T1 to T2.                      |
| <b>Spot-next</b>   | A deposit or FX swap <b>from spot until the “next” day, or day after spot</b> — T2 to T3.           |
| <b>Spot-a-week</b> | A deposit or FX swap <b>from spot until a week later</b> — T2 to T7 from T2, or T9 from T0.         |

In dealing-room speak these are referred to as **value today, value tom, spot-next and spot-a-week**.

### §1.7.3 Forward-forwards

A swap deal between two forward dates rather than from spot to a forward date — e.g. **sell USD in the 1-month and buy them back in the 3-month**, giving a “**net swap position**” for two months starting in one month. A company might do this because it has an FEC but **wishes to defer its utilisation by a further two months**. From the bank's point of view a forward-forward swap is **two separate swaps, each based on spot** (also referred to as a spot swap).

### §1.7.4 The three types of FEC

At times it is difficult for clients to anticipate the **exact date** on which a payment will be made (imports) or received (exports), so the client can select:

**Fixed Contract** — with a fixed delivery date. **OR**

**Part Optional Contract** — an option to deliver **at any time between 2 dates**. **OR**

**Fully Optional Contract** — an option to deliver **at any time on or before the delivery date**.

### The mechanics

- **The settlement of local trade transactions is not that difficult**, as both parties are dealing in South African rand (ZAR)
- **Foreign exchange happens when there is a need to change one currency for another country's currency using the banking system** — the case when buyers and sellers in different countries trade with one another across borders
- **Both sellers (exporters) and buyers (importers) have conflicting requirements:** “if I ship my goods, will I be paid?” and “if I pay, will I receive my goods?”
- The **seller wants to optimise DSO — days sales outstanding** — while the **buyer would like to pay as late as possible — DPO, days payable outstanding**
- There are normally **2 parties to a trade transaction, the seller and the buyer**. Their **local banks act as intermediary parties and are designated “Authorised Dealers”**, with foreign exchange departments that handle international payments on behalf of their customers
- **The primary role of a bank is to provide risk reduction alternatives to both the buyer and the seller**
- Transactions are settled using a **secured telecommunication network called SWIFT — Society for World-wide Interbank Financial Telecommunications** — using **authenticated SWIFT addresses**

**TRADE TRANSACTIONS, START TO FINISH** Trade transactions begin with the negotiation of a contract between a buyer and a seller and end with the exchange of goods or services for monetary payment.

**EXAM WATCH · THE EXCHANGE CONTROL RULES** “South African residents are not permitted to speculate against the Rand.” “South African residents are bound by South African exchange control laws.” In terms of the Exchange Control Laws **only Authorised Dealers (ADs) and Authorised Dealers with Limited Authority (ADLAs)**, appointed by the South African Reserve Bank (SARB), are permitted to buy, sell, borrow or lend foreign currency. South African individuals and entities must **convert ZAR to foreign currency to effect cross-border payments for imported goods or services**, or **convert foreign currency to ZAR when receiving cross-border funds** — done by using an exchange rate to convert.

### Seven factors to keep in mind on any cross-border trade

The relationship between the seller and buyer · the payment and shipment risks (insurance) · currency risk (fluctuation of exchange rates) · foreign exchange risk — transaction exposure, translation exposure and economic/operating exposure · the type of goods to be imported — is it permissible or not, as per SARB requirements · political/economic/sanctions situation in both countries (trade restrictions) · the timing of cash flow and delivery, referring to the **basic operating cycle**: cash → purchase of goods or raw material → process goods or service for sale → sell goods or services → collect cash from customer → cash.

## §1.9 · INTERNATIONAL PAYMENT METHODS AND THEIR RELATIVE RISK

The main international payment methods used around the world today are **Cash in Advance · Open Account · Documentary Collections · Documentary Credit (L/C)**.

| METHOD                 | RISK TO SELLER | RISK TO BUYER | WHY                                                                                                                                                                                                                                                           |
|------------------------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Payment in advance     | VERY LOW       | VERY HIGH     | Carries the <b>lowest risk for the seller</b> as he will <b>receive most or all of the funds before shipment</b> , but the <b>highest risk for the buyer</b> as <b>goods may never be shipped or incorrect goods received</b> .                               |
| Open account           | VERY HIGH      | VERY LOW      | Highest risk for the seller — <b>no guarantee the buyer will be willing and able to pay on the due date</b> , and the <b>buyer will be in possession of the goods before payment</b> .                                                                        |
| Documentary collection | MEDIUM         | MEDIUM        | Collections <b>offer some protection to both seller and buyer</b> , in that the <b>mechanism for exchange of documents and payment is arranged via the banking system</b> .                                                                                   |
| Documentary credit     | MEDIUM         | MEDIUM        | As <b>banks only deal in documents</b> , the <b>importer must pay if the documents conform</b> , even if it is the <b>incorrect goods</b> ; and the <b>exporter might not get paid if the documents are discrepant</b> , as the <b>guarantee falls away</b> . |

**EXAM WATCH** The two extremes are **mirror images**: **payment in advance is very low risk to the seller and very high to the buyer; open account is exactly the reverse**. The two documentary methods sit in the middle for **both parties**, because the **banking system intermediates the exchange of documents against payment**.

**THE DOCUMENTARY-CREDIT TRAP** **Banks only deal in documents, not goods**. That single fact produces both risks: the **importer must pay on conforming documents even if the goods are wrong**, and the **exporter loses the L/C guarantee the moment his documents are discrepant**. Neither party's risk is about the merchandise — it is about the paperwork.

## §1.9 · COMPARISON OF PAYMENT TERMS WITH RELATIVE RISK

| METHOD OF PAYMENT                                              | DOCUMENTS AVAILABLE TO BUYER                              | TIMING OF PAYMENT TERMS TO SELLER                                                  | RISK TO BUYER                                                                            | RISK TO SELLER                                                                                        |
|----------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Payment in advance</b>                                      | After payment                                             | Before shipment of goods                                                           | <b>Non-delivery of goods</b>                                                             | <b>Could lose the sale to competitors</b> , due to payment terms                                      |
| <b>Open account payment</b>                                    | Upon delivery                                             | Upon payment of invoice                                                            | <b>No risk</b>                                                                           | <b>Full reliance on the willingness of the buyer to pay</b>                                           |
| <b>Documentary collections — D/P</b><br>documents @ payment    | After payment                                             | Upon sighting & presentation of documents by the bank                              | <b>Payment before goods are cleared through their customs</b>                            | The <b>buyer could decide not to take up documents</b> and the goods need to be returned (no payment) |
| <b>Documentary collections — D/A</b><br>documents @ acceptance | Upon acceptance to pay at a later fixed determinable date | Upon maturity of term payment                                                      | <b>Legal actions for non-payment</b> if protesting was stipulated                        | —                                                                                                     |
| <b>Sight documentary credit</b>                                | Upon settlement of the L/C                                | Once complying documents have been presented to the negotiating/paying bank        | <b>Goods are not correct as described in documents — the buyer's bank will still pay</b> | <b>Very little risk unless documents are discrepant</b> — the L/C guarantee will then fall away       |
| <b>Term documentary credit</b>                                 | Upon the bank releasing documents of title                | Upon maturity of the L/C terms, or the discounting of the L/C by the seller's bank | <b>Payment is due on a fixed date even if there is a dispute about the goods</b>         | <b>Very little risk unless documents are discrepant</b> — the L/C guarantee will then fall away       |

**EXAM WATCH** Note the asymmetry the table encodes: under **open account the buyer carries no risk at all**, and under a **documentary credit the buyer's risk is that the goods are wrong but the bank pays anyway**. In both documentary credit rows the seller's risk is **identical and document-driven** — discrepant documents, guarantee gone.

## §1.10 · FOREIGN EXCHANGE RISKS

**Foreign exchange risk** — also known as **exchange rate risk**, the **risk of financial impact due to exchange rate fluctuations**. In simpler terms, the risk that a **business' financial performance or financial position will be impacted by changes in the exchange rates between currencies**.

| RISK                                                 | DEFINITION                                                                                                                                                                                  |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economic risk</b><br>also known as operating risk | The impact of currency fluctuations on a company's <b>profitability, market share and competitive position</b> .                                                                            |
| <b>Transaction risk</b>                              | Increases when there is a <b>time difference between entering and settling a contract</b> . This allows exchange risks to fluctuate, which can lead to <b>increased transaction costs</b> . |
| <b>Translation risk</b>                              | The risk that a company's <b>assets, liabilities, income or equities will change in value due to exchange rate changes</b> .                                                                |
| <b>Jurisdiction risk</b>                             | When <b>laws unexpectedly change in the country where the exporter is doing business</b> . <b>Less common and primarily exists in unstable countries</b> .                                  |
| <b>Currency risk</b>                                 | The risk traders assume when they <b>make transactions in foreign currency</b> — businesses <b>can lose value if they hold a weakening currency</b> .                                       |
| <b>Interest rate risk</b>                            | <b>When interest rates attract foreign capital, the foreign exchange rate will rise. When interest rates decrease, exchange rates will also drop.</b>                                       |
| <b>Currency restrictions</b>                         | Some countries <b>impose restrictions on currency conversion and expatriation</b> , which can make it <b>difficult to access and repatriate earnings</b> .                                  |

### §1.10.1 The three exposures

| EXPOSURE                             | DEFINITION                                                                                                                                                                                                                                                        |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Transaction exposure</b>          | The degree to which <b>future FX-denominated cash flows from contractually binding transactions are affected by currency fluctuations</b> .                                                                                                                       |
| <b>Translation exposure</b>          | The risk that a company's <b>equities, assets, liabilities or income will change in value because of exchange rate changes</b> .                                                                                                                                  |
| <b>Economic / operating exposure</b> | A measure of the <b>change in the future cash flows of a company as a result of unexpected changes in foreign exchange rates</b> . Economic exposure cannot be easily mitigated, because it is caused by the unpredictable volatility of currency exchange rates. |

**EXAM WATCH** The **three exposures map onto three of the risks** — transaction, translation and economic — so learn them as one set, not two. The distinguishing line: **transaction exposure is about contracted cash flows, translation exposure is about the balance sheet, and economic exposure is about future cash flows and cannot be easily hedged**.

## The formulas

| QUANTITY                 | FORMULA                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------|
| Value of one point (pip) | Size of deal × Net change in rate — <b>expressed in the base currency</b>                         |
| Reciprocal rate          | $1 \div$ the quoted rate — <b>and the two sides of the quotation reverse</b>                      |
| Purchasing power parity  | Cost of the basket in country A $\div$ cost of the same basket in country B                       |
| Spread                   | Offer – Bid — <b>a risk spread, not a dealer profit</b>                                           |
| Forward rate             | Spot adjusted for the <b>interest rate differential</b> between the two currencies for the period |

## The directions

| EVENT                       | RAND               | CONSEQUENCE                                                                                                        |
|-----------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------|
| USD/ZAR rises (7.11 → 7.20) | ZAR<br>depreciates | Import prices <b>increase</b> , inflation <b>increases</b> , exports <b>increase</b> , employment <b>increases</b> |
| USD/ZAR falls (7.07 → 6.98) | ZAR<br>appreciates | Import prices <b>decrease</b> , inflation <b>decreases</b> , exports <b>decrease</b> , employment <b>decreases</b> |
| SARB <b>buys</b> USD        | ZAR<br>depreciates | M1 <b>increases</b> , rand interest rates <b>decrease</b>                                                          |
| SARB <b>sells</b> USD       | ZAR<br>appreciates | M1 <b>decreases</b> , rand interest rates <b>increase</b>                                                          |

## APPENDIX · THE TRAPS

| TRAP                      | THE RIGHT ANSWER                                                                                                                                                             |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Whose point of view?      | Exchange rates are always quoted from the bank's point of view. The bank <b>buys the base low on the left and sells it high on the right</b> .                               |
| The spread                | <b>Does not represent a dealer profit but a "risk spread"</b> .                                                                                                              |
| Spot settlement           | Most spot markets settle <b>one</b> business day after the trade, but <b>foreign exchange settles in two</b> .                                                               |
| Reciprocal sides          | The <b>smaller number is always on the left</b> , so <b>the two sides reverse</b> when you invert a quotation.                                                               |
| Which is base?            | By convention <b>EUR, GBP, AUD and NZD are quoted as the base</b> against the USD; almost everything else has the <b>USD as base</b> .                                       |
| Cross rate                | Any quotation between two currencies <b>where the US dollar is not one of them</b> . In SA <b>all rates are quoted against the USD and everything else is a cross rate</b> . |
| Premium and discount      | <b>PIL · PEG · DIG · DEL</b> . A <b>premium benefits the exporter and costs the importer</b> ; a discount does the reverse.                                                  |
| Forwards are not free     | A forward <b>guards against adverse moves but also precludes profit from favourable moves</b> .                                                                              |
| Payment in advance        | <b>Very low risk to the seller, very high to the buyer</b> — open account is exactly the reverse.                                                                            |
| Documentary credit        | <b>Banks only deal in documents</b> — the importer must pay on conforming documents <b>even if the goods are wrong</b> .                                                     |
| Who may deal in FX in SA? | <b>Only ADs and ADLAs appointed by the SARB. SA residents may not speculate against the rand</b> . The <b>FSCA</b> is the market conduct regulator.                          |
| Optimal currency value    | <b>There is no optimal value for a currency</b> — a strong currency helps some sectors and hurts others.                                                                     |

**SOURCES** FMP V05 Mod07 FX Markets STUDY GUIDE 25 10 03 CJ (27pp) and FMP V06 Mod07 FX Markets CLASS Pres 26 09 02 CJ (65 slides). Section numbers used throughout are the guide's own. Question 3 of the workbook additionally requires the **SARB Currency and Exchange Manual for Authorised Dealers**, downloaded from the SARB website.